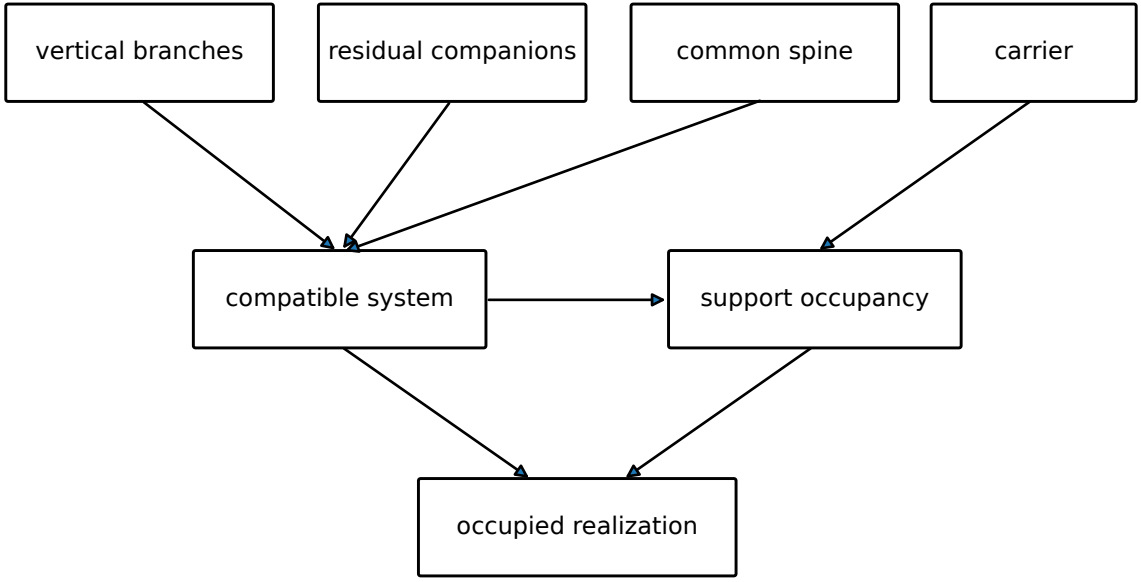the conclusion requires all four input blocks